Arushi Taneja

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Education

University of Wisconsin - Madison Madison, WI
B.S. Computer Science and Data Science *Anticipated graduation May 2026*

- **GPA:** 3.541/4.000
- **Relevant Coursework:** Algorithms, Operating Systems, Java Programming, Data Science Programming, Computer Engineering, Discrete Mathematics, Machine Organization, Data Science Modeling, Building User Interfaces, Computer Graphics, Software Engineering, Artificial Intelligence, Database Systems

Experience

Data Science Programming II Peer Mentor | *Python, Git, UNIX, Flask, Selenium* March 2025 – Present
University of Wisconsin - Madison *Madison, WI*

- Mentor 300+ students in Python programming and debugging, improving assignment completion rates and confidence with data structures.
- Guide students through problem-solving and A/B testing strategies, strengthening analytical thinking and code robustness in data-centric workflows.
- Standardize workflows for Git and virtual environments, increasing reproducibility and version-control adoption across the course.
- Foster an inclusive learning environment by tailoring technical explanations to diverse learning styles and skill levels.

Software Developer | *C#, UWP, .NET, Azure, MSSQL, MVVM* June 2025 – September 2025
365Labs *Baton Rouge, LA*

- Engineered new features for the Records Management System to meet multi-state IBR/NIBRS compliance, ensuring 100% data validity across jurisdictions.
- Designed an automated pipeline for PDF extraction and auto-fill using iText7, GdPicture OCR, PSPDFKit, and OpenAI, cutting manual entry time by 60%.
- Built a JSON mapping engine for heuristic field matching, reducing template creation and boosting pipeline flexibility.
- Implemented unit testing frameworks for state-specific compliance rules, increasing module reliability and reducing defects before deployment.

Undergraduate Research Assistant | *Javascript, ReactNative, Firebase* September 2024 – May 2025
Magic Lab,University of Wisconsin - Madison *Madison, WI*

- Re-architected real-time motion-capture integration with Firebase to minimize crash frequency and improve data consistency by 40%.
- Developed a custom tween function for fluid pose transitions, enhancing frame-stability and interaction smoothness in gesture models.
- Built a dynamic script-editor module enabling researchers to prototype interactive narratives without runtime failures.
- Collaborated in Scrum sprints with developers and researchers, delivering iterative accessibility improvements and system stability updates.

Projects

Qualcomm Capstone Project | *Python, Computer Vision, Edge AI, PyTorch* September 2025 – Present
AI-Powered Feedback Coach

- Develop an AI tool on Snapdragon PCs for real-time posture, gesture, and gaze analysis.
- Collect and label a multimodal dataset to train pose and gaze estimation models, improving engagement-score prediction accuracy by 25%.
- Optimize the video inference pipeline with Qualcomm AI SDKs (QAIRT, ONNX Runtime, LiteRT) to achieve sub-second latency and smooth visualization.

Driver's Application | *Javascript, ReactNative, Expo, Azure, MSSQL* December 2024 – Present

- Lead the design and development of a real-time task-management app for truck drivers to streamline proof-of-delivery tracking and communication.
- Build a full-stack solution with a React Native frontend, RESTful API backend, and MSSQL database on Azure to ensure scalable, reliable performance.
- Integrate secure document uploads via Azure Blob Storage, improving proof-of-delivery reliability and reducing data-loss incidents by 40%.

Technical Skills

Programming Languages: C, C++, Java, Python, C#, JavaScript, SQL, R, Assembly, HTML/CSS
Frameworks & Libraries: .NET, React, React Native, PyTorch, SpringBoot, Pandas, NumPy, MediaPipe, JUnit, Bootstrap
Tools & Platforms: Azure, Google Cloud Platform, Microsoft SQL Server, Git, Docker, VS Code, Visual Studio, IntelliJ
Core Strengths: Algorithms, distributed systems, scalable infrastructure, relational databases, performance optimization, debugging, test-driven development, cloud-based infrastructure, version control & CI/CD, Agile & Scrum Development